

Python Cheat Sheet - Coding Round Essentials

Basic Input/Output

```
x = int(input())
a, b = map(int, input().split())
arr = list(map(int, input().split()))
```

Strings

```
s = "hello"
s[::-1]
s.count('l')
''.join(list_of_chars)
''.join(sorted(s))
s == s[::-1]
```

Lists

```
arr = [1, 2, 3]
arr.append(4)
arr.pop()
arr.sort()
arr.reverse()
arr.remove(2)
arr[1:3] # [2, 3]
```

Dictionary (HashMap)

```
d = {'a': 1, 'b': 2}
d['a']
d.get('c', 0)
for key, value in d.items():
    print(key, value)
```

Sets

```
s = set([1, 2, 3])
s.add(4)
s.remove(1)
s1 | s2
s1 & s2
```

Loops and Conditions

```
for i in range(5):
    print(i)
while x > 0:
    x -= 1
if a > b:
    ...
elif a == b:
    ...
else:
    ...
```

Functions & Recursion

```
def fact(n):
    if n == 0:
```

```
    return 1
return n * fact(n-1)
```

Useful Libraries

```
from collections import Counter, deque
Counter([1, 2, 2, 3])
q = deque()
q.append(1)
q.popleft()
```

Tricks

```
a, b = b, a
squares = [x*x for x in range(10)]
all(x == arr[0] for x in arr)
max(set(arr), key=arr.count)
```

Problem-Solving Templates

Two Sum

```
def two_sum(nums, target):
    d = {}
    for i, val in enumerate(nums):
        if target - val in d:
            return [d[target - val], i]
        d[val] = i
```

Palindrome Check

```
def is_palindrome(s):
    return s == s[::-1]
```

Frequency Counter

```
freq = Counter(arr)
```

BFS (Graph)

```
from collections import deque
def bfs(graph, start):
    visited = set()
    q = deque([start])
    while q:
        node = q.popleft()
        if node not in visited:
            visited.add(node)
            q.extend(graph[node])
```